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PALEOMAGNETIC RESULTS FROM THE SHASTA BALLY PLUTONIC BELT IN THE KLAMATH MOUNTAINS PROVINCE, NORTHERN CALIFORNIA

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Abstract. Available paleomagnetic data show approximately 100° of clockwise rotation for Permian and Triassic strata of the Eastern Klamath terrane. Jurassic strata of this terrane are rotated approximately 60° clockwise, which is comparable to rotations reported for Jurassic plutons that occur elsewhere in the Klamath Mountains province. Paleomagnetic data obtained during the present study from the Shasta Bally belt of Cretaceous plutons indicate 25.7° ± 13.6° of clockwise rotation for the province since Early Cretaceous time (≃ 136 Ma). The waning stages of rotation at the time of emplacement of the Shasta Bally belt plutons, which was closely followed by deposition of basal strata (Lower Cretaceous) of the Great Valley sequence, probably represents completion of accretion of the province to cratonic North America.

Introduction

The Klamath Mountains province of northern California and southern Oregon consists of a series of ophiolitic fragments and volcanic island arcs that formed during Paleozoic and Mesozoic times. Available paleomagnetic evidence indicates that some Devonian to Jurassic rocks of the Klamath Mountains have undergone large rotations relative to a stable North America since their formation [Achache et al., 1982; Fagin and Gose, 1983; Schultz, 1983; Mankinen et al., 1984; Bogen, 1986]. A paleomagnetic study of superjacent Cretaceous strata of the Klamath Mountains [Mankinen and Irwin, 1982] indicated that little or no rotation of the province had occurred since Cretaceous time and perhaps not since Early Cretaceous. No paleomagnetic evidence for significant latitudinal displacements has been found for any of the Klamath terranes.

In order to determine better when rotation of the Jurassic and older rocks ceased, we obtained paleomagnetic samples from the Shasta Bally plutonic belt in the southern Klamath Mountains. This belt consists of Cretaceous granitoid rocks that are generally the youngest (≃ 136 Ma) plutons of the province [Irwin, 1984]. Plutons of the Shasta Bally belt intrude mainly the Eastern Klamath and Central Metamorphic terranes. At its northern end, the belt is partly superimposed on the oldest plutonic belt (Alpine gabbro belt) which is Ordovician in age. These and other relations indicate that the plutons of the Shasta Bally belt were emplaced after the various terranes that constitute the province joined together (amalgamated), and probably during or shortly after accretion of the Klamath terranes to the North American continent.

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Methods and Results

A common difficulty in the interpretation of paleomagnetic results from plutonic rocks is that the rocks often lack the structural control necessary to detect post-emplacement tilting. Even though structural control is lacking for many of the Klamath Mountain plutons, we proceeded under the assumption that tilting of the plutons of the Shasta Bally belt has been negligible. The fact that results from some of the older plutons that were sampled by Schultz [1983] could, in general, be interpreted by assuming only vertical-axis rotation allowed us to have some confidence in our basic premise. Paleomagnetic samples were obtained from the Craggy Peak and Sugar Pine plutons and the Shasta Bally batholith, all of which intrude the Eastern Klamath terrane. We were unable to sample some plutons of the belt because of poor exposures due to deep weathering, and others because they are located in primitive areas.

A total of 82 oriented cores was obtained from 9 sites shown in Figure 1. Site 1 is within one of the several small satellitic plutons east of the Shasta Bally batholith [Kinkel et al., 1956; Albers, 1964], and a number of dikes in the metamorphic aureole of the Shasta Bally batholith [Irwin, 1963] were sampled at site 6. Remanent magnetization was measured using a cryogenic magnetometer. Stability of remanence was tested by alternating field (AF) demagnetization using a commercial tumbling demagnetizer, and by thermal demagnetization using a magnetically shielded oven with an ambient field of less than 10 Gammas. Both methods had a comparable effect in removing anomalous components of magnetization (Figure 2) and AF demagnetization was used for the majority of samples. Because of a large variability in magnetic stability, however, all samples were progressively demagnetized and characteristic magnetizations were chosen individually. Site-mean remanence directions, after magnetic cleaning, are shown in Figure 3, and the means and associated statistical parameters are given in Table 1.

Both normal- and reversed-polarity sites were sampled in the Shasta Bally batholith (Figure 3), and the mean remanence directions are antipodal. This positive reversal test is indicative of a primary magnetization of the batholith. The anomalous magnetization directions for Shasta Bally sites 5 and 6 indicate that these rocks may have acquired their remanence while the geomagnetic field was undergoing a reversal of polarity, or that the sites were tilted and perhaps rotated. Undetected tilting (with or without accompanying vertical-axis rotation) is possible

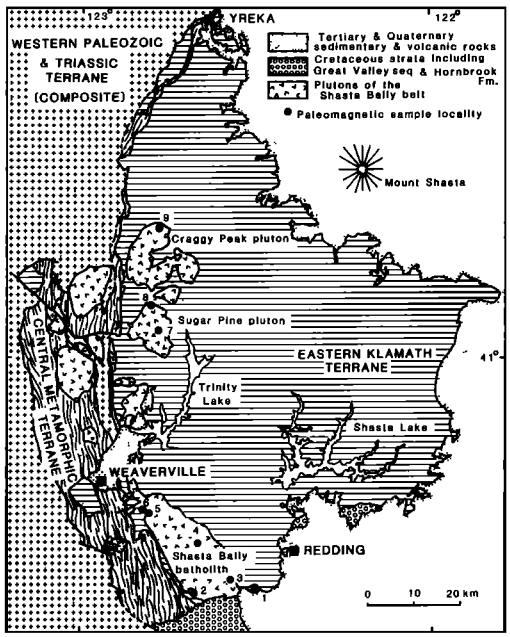


Fig. 1. Sketch of a portion of the Klamath Mountains province showing major plutons of the Shasta Bally belt and sample localities.

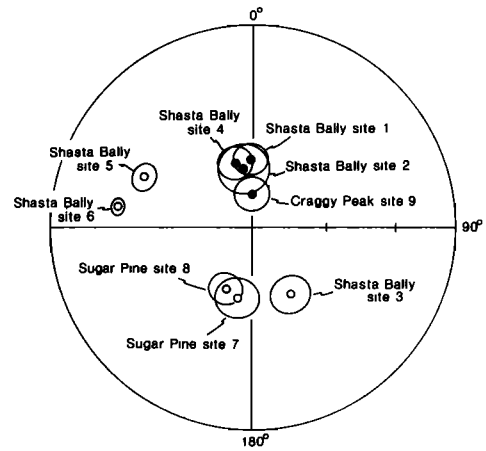


Fig. 3. Average remanent magnetization directions for all sites after magnetic cleaning. Ovals, 95% confidence limits about mean directions. Solid symbols, directions on lower hemisphere of equal-area projection; open symbols, directions on upper hemisphere.

because this area of exposure is topographically separate from the main mass of the batholith. Which of the two alternatives best explains the unusual direction of these sites is unknown, but it is clear that their remanence direction is not representative of the geomagnetic field at the time of emplacement of the batholith.

The Craggy Peak pluton, which is near the northern end of the plutonic belt, yields a normal-polarity direction that agrees well with sites within the Shasta Bally batholith (Figure 3). The two reversed-polarity sites within the Sugar Pine pluton, however, have remanence directions that are somewhat discordant with those of the other two plutons (Figure 3). This discordance could be indicative of a small amount of tectonic disruption of this pluton relative to the other two sampled; as little as 15° of tilt toward the southwest could produce this direction. Alternatively, the differences in remanence direction seen in Figure 3 (exclusive of sites 5 and 6) could be a reflection of the true dispersion within the plutonic belt.

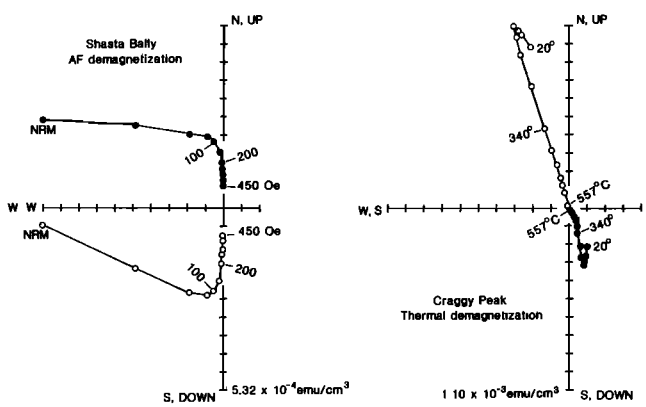


Fig. 2. Orthogonal projections of remanence vector during alternating field and thermal demagnetization. Solid symbols, projections into the horizontal plane; open symbols, projections into the vertical plane.

Discussion and Conclusions

Achache et al. [1982], Fagin and Gose [1983], and Schultz [1983] considered the discordant directions indicated by their paleomagnetic data to be the result of rigid-body rotation about vertical axes during Tertiary time in concert with rotation of the Oregon Coast Ranges. Achache et al. [1982], however, assumed that remagnetization of the limestone in the Devonian Kennett Formation was caused by the intrusion of the Shasta Bally batholith. Because the remanence directions of the batholith and the directions recorded by the Kennett are markedly dissimilar, the batholith could not have been the cause of the remagnetization. This remagnetization probably occurred during Jurassic time [Mankinen et al., 1984] and rotation could, therefore, have occurred much earlier than suggested by Achache et al. [1982]. Bogen [1986] considered the available paleomagnetic data to indicate either rigid-block rotation of two separate areas of the Klamath Mountains, or rigid-block rotation of the entire province accompanied by oroclinal bending of the northwestern region. This rotation was considered to be post-Late Jurassic. Mankinen and Irwin [1982] and Mankinen et al. [1984] also assumed rigid-block rotation but showed that it occurred during Mesozoic time and that, therefore, significant rotations in the Klamath Mountains occurred independently of the Oregon Coast Ranges. Contrary to these interpretations, Renne et al. [1986a, 1986b] consider that their unpublished paleomagnetic data from the Eastern Klamath terrane, when compared with those previously published, are indicative of vertical-axis oroclinal deformation. However, these conclusions cannot be properly assessed until the data become avail-

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TABLE 1. Site-Average Paleomagnetic Data for Plutons of the Shasta Bally belt, Klamath Mountains

Unit	Site Number	Site Lat.	Site Long.	N/N_o	Treatment	Remanence Direction Incl.	Decl.	k	α_{95}	J_{cl}	Paleomagnetic Pole Lat.	Long.	A_{95}
Shasta Bally batholith	1	40.49	237.50	9/9	AF+T	62.5	358.7	63	6.5	1.410	86.5	222.0	9.2
	2	40.49	237.33	7/8	AF+T	66.1	351.5	32	10.8	0.417	80.0	203.0	16.5
	3	40.54	237.43	8/8	AF+T	-58.9	149.9	54	7.6	1.190	-67.0	335.0	10.1
	4	40.61	237.34	7/10	AF	63.3	345.5	73	7.1	3.230	78.6	173.6	10.2
	5*	40.66	237.20	10/10	AF	-40.7	295.1	97	4.9	2.010	02.2	113.5	5.0
	6*	40.67	237.18	10/10	AF	-33.4	278.9	279	2.9	19.200	-05.3	127.7	2.7
		Shasta Bally batholith mean direction (N = 4 sites)				63.1	345.8	156	7.4		78.9	174.1	11.0
Sugar Pine pluton	7	41.07	237.22	10/10	AF+T	-61.0	191.5	34	8.5	2.730	-81.4	136.6	11.7
	8	41.11	237.20	6/7	AF+T	-63.1	202.5	110	6.4	4.970	-73.2	127.8	9.1
		Sugar Pine pluton mean direction (N = 16 samples)				-61.9	195.5	45	5.5		-78.4	131.9	7.8
Craggy Peak pluton	9	41.28	237.22	10/10	AF+T	76.7	359.0	52	6.8	6.320	66.5	236.1	12.4
		Shasta Bally plutonic belt mean direction (N = 7 sites)				65.5	356.5	71	7.2		82.4	220.1	11.0
		Shasta Bally plutonic belt mean direction (N = 5 sites)				65.9	347.3	91	8.1		78.1	194.5	11.9

Lat., Long.= north latitude and east longitude, respectively; N/N_o = number of samples used in the analysis/total number of samples collected; Treatment= alternating field (AF) or thermal (T) demagnetization; Incl.= inclination, positive downwards; Decl.= declination, east of north; k = precision parameter (N-1/N-R); α_{95} (A_{95}) = 95% confidence cone about average direction (pole); J_{cl}= geometric mean intensity after magnetic cleaning (x 10^{-5} emu/cm^3).

*Sites excluded from calculation of mean direction of batholith (see text).

able. We believe that all published results, in general, can be most easily interpreted in terms of rigid-block rotation.

Most of the paleomagnetic information concerning rotation of the Klamath Mountain terranes comes from the Redding section of the Eastern Klamath terrane. The pre-Permian paleomagnetic history, however, is unknown. The similarity of indicated clockwise rotation for Permian and Triassic strata of the Eastern Klamath terrane (≈ 100°) shows that major rotation of these rocks began in latest Triassic or earliest Jurassic time [Mankinen et al., 1984]. This rotation is logically assumed to have included pre-Permian strata and preamalgamation plutons of the terrane, together with the Central Metamorphic terrane which joined the Eastern Klamath nucleus of the province during Devonian time [Irwin, 1981]. The youngest Jurassic strata sampled in the Eastern Klamath terrane indicate clockwise rotation of approximately 60° [Mankinen et al., 1984]. Paleomagnetic samples have been obtained from the Ashland, Grants Pass, Greyback, Slinkard, and Wooley Creek plutons in the composite Western Paleozoic and Triassic terrane [Schultz, 1983]. These Late Jurassic plutons, with the exception of the Ashland pluton, yield rotations that are comparable to those recorded by the Jurassic strata in the Eastern Klamath terrane if only vertical-axis rotation is assumed. The agreement of paleomagnetic data between the two areas suggests that the Western Paleozoic and Triassic terrane, the Eastern Klamath terrane, and the intervening Central Metamorphic terrane were consolidated by Late Jurassic and have rotated as a single block since that time. Indeed, the suturing of the individual terranes within the Western Paleozoic and Triassic terrane, and the composite terrane to the Central Metamorphic terrane have been considered to be mid-Jurassic events on the basis of various geologic and geochronologic criteria [Irwin, 1981; 1984].

Comparison of the mean pole that includes the reliable sites from all three plutons (82.4°N., 220.1°E., A_{95} = 11.0°) with a Cretaceous reference pole [Gromme, 1984] yields a clockwise rotation of 25.7° ± 13.6°. (Uncertainties are 95% confidence limits calculated using the method of Demarest [1983]). No significant latitudinal displacement (northward 4.9° ± 9.0°) is indicated by these data, as is true for all previous studies in the province. If one assumes that the Sugar Pine pluton has been disrupted and excludes sites 7 and 8 from the mean, the indicated clockwise rotation is reduced to 16.7° ± 14.8°. Any rotation of the Shasta Bally plutonic belt must apply to the entire Klamath Mountains province because the last element added to the province, the Western Jurassic terrane, was in place by Late Jurassic time [Irwin, 1981]. Because the isotopic age of the Shasta Bally batholith (136 Ma) is remarkably close to the Early Cretaceous (Valanginian) age of the basal strata of the onlapping Great Valley sequence, the age difference between them is unlikely to be more than a few million years. The close correspondence between the last stages of rotation and beginning deposition of basal strata of the Great Valley sequence on Klamath basement probably represents completion of accretion of the Klamath Mountains to the North American continent.

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Similar rotations are seen when comparing results from the Shasta Bally plutonic belt, the Cretaceous onlap sequences (Great Valley sequence and Hornbrook Formation), and Tertiary volcanic rocks in part of the Cascade Range. The superjacent Cretaceous strata studied by Mankinen and Irwin [1982] indicate 14.4° ± 6.9° of clockwise rotation when compared with Grommé's [1984] Cretaceous reference pole. This rotation differs from that previously reported because here we have converted the data from field directions to their corresponding virtual geomagnetic poles to conform with current practice. Both remagnetized (in-situ) and unremagnetized (structurally corrected) data sets also were combined before comparing the mean with the reference pole. Volcanic rocks in the southern Cascade Range immediately adjoining the Klamath Mountains province to the east have yielded a clockwise rotation of 14.0° ± 9.0° [Beck et al., 1986]; these two rotations are of the same magnitude as the 25.7° ± 13.6° of clockwise rotation reported herein for the Shasta Bally plutonic belt.

The similar rotations indicated by the different rock types in the three areas are unlikely to be fortuitous, but instead are strongly indicative that the southern Klamath Mountains province, the northern Great Valley sequence, and the southern Cascade Range rotated as a single rigid block during Tertiary time. Much of the rotation seen in the southern Cascade Range rocks occurred during the late Oligocene [Beck et al., 1986], so the last 20° or so of rotation indicated by the rocks of the Klamath Mountains seems unlikely to have been part of the accretionary process but was probably related to the earliest stages of extension in the Basin and Range province. The similarity of rotation of the plutons to that of the Cretaceous and Tertiary strata thus indicates that the plutons of the Shasta Bally belt probably are postaccretion as well as postamalgamation.

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